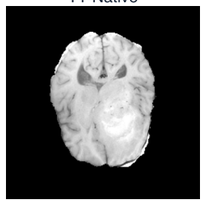
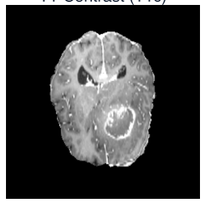
(a) Multi-Modal 3D MRI

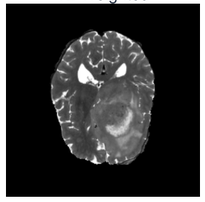
T1-Native



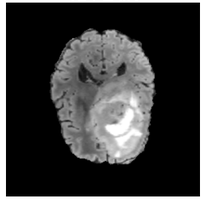
T1-Contrast (T1c)



T2-Weighted

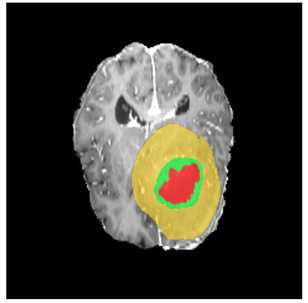


T2-FLAIR



(b) 3D Segmentation (ROI)

SegResNetB0 (92.2% WT)



☐ Necrotic Core (NCR)

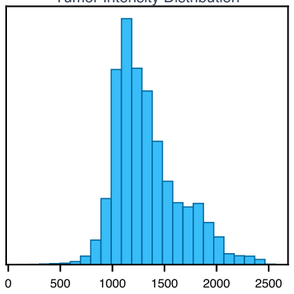
☐ Peritumoral Edema (ED)

☐ Enhancing Rim (ET)

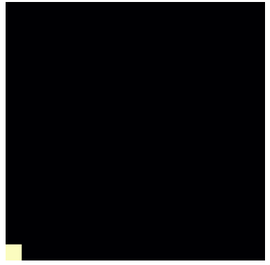
WT: 136.9 cm³ | TC: 31.0 cm³

(c) Radiomic Extraction

Tumor Intensity Distribution



FLAIR GLCM Texture Matrix

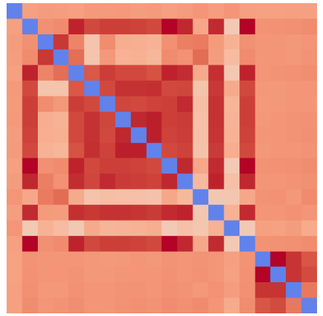


3D Morphometry

Flatness: 0.587
MeshVolume: 136.9 cm³
LeastAxis: 34.2 mm
Skewness: 0.418

(d) Quantum QUBO Pruning

20×20 QUBO Hamiltonian

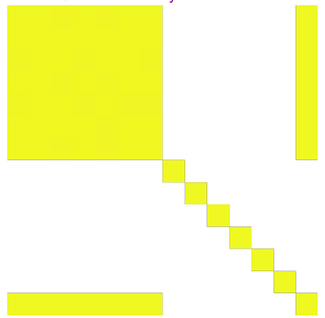


IBM Fez (156 QPU)
10,000 Physical Shots
Ground E = -3.0981

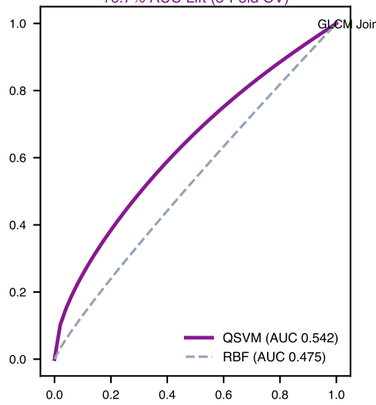
9 Selected Biomarkers
(Jaccard J = 0.491)

(e) 9-Qubit QSVM (512D)

Quantum Fidelity Gram Matrix

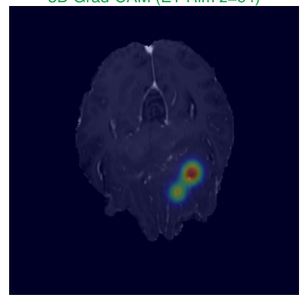


+6.7% AUC Lift (5-Fold CV)

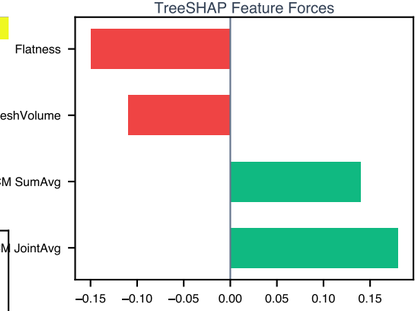


(f) Clinical Decision & XAI

3D Grad-CAM (ET Rim z=64)



TreeSHAP Feature Forces



MGMT: Unmethylated (57.0%)
IDH1/2: Wildtype (81.8%)
Risk: High / Aggressive Course
TMZ Resp: Low / Moderate